P2

PLANT OPERATION MODEL "POM"

Cross Functional Process

LAFARGE EMIRATES CEMENT

Instrumentation Quality Assurance

What is a Critical Sensor?

- Sensor is critical to run the plant without affecting safety, environment, quality and performance.
- Sensor that guide operators to achieve target productivity, quality, and reliability.
- Sensor that can affect Safety (prevent explosion and fire).
- Sensor measure indicators essential for legal compliance.
- Sensor measure indicator essential for customer satisfaction.
- Sensor is 100% necessary to operate and make appropriate operation decisions.

What are the consequences of critical Sensor failure?

- High risk on safety to operating personnel at site.
- Product Quality is at high risk (customer complaint).
- Legal compliance is at high risk.
- Main equipment breakdown causing Reliability issue.
- Productivity is at high risk and wrong decisions.

What is Validation?

- Validation is a parallel measurement to the plant installed probe and measurement chain.
- Objective is to cross check (where possible) the central control room reading with a suitable portable instrument.
- Validation could also be achieved with a logical check between linked parameter.
- Validation master plan is a schedule to cross check all critical sensor frequently.
- Implementation of defined validation master plan should be monitored.
- Validation results should be documented, reviewed and analyzed.

Instrumentation Quality Assurance

What is Calibration?

- Objective of calibration is to check the accuracy of the instrument, normally following a supplier's methodology under specified conditions.

- The calibration should correct drifts of the instrument itself but does not validate that the complete measuring chain is operating correctly nor the measuring point is representative of the process.

- Calibration master plan is a schedule to calibrate all critical sensor frequently.

- Implementation of defined calibration master plan should be monitored.

- Calibration results should be documented, reviewed and analyzed.

Communication Benefits

The person who has carried out the validation or calibration has to inform the end user and all concern departmets.

This is key to:

- Avoid wrong reactions from system / operator

- Update impacted targets